

José Luis La Torre Romero

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Final-year Software Engineering student with experience in the development of IoT systems and Artificial Intelligence solutions. Hands-on experience developing prototypes with ESP32, computer vision, and Machine Learning, integrating physical devices with backend services through REST APIs. Additional experience in Python, C/C++, TensorFlow, and distributed systems development.

PROJECTS

[SmartTachito](#)

IoT System for Automatic Waste Classification Using Computer Vision

github.com/adminLTR/SmartTachito

- Trained a **MobileNet CNN** model for waste classification using **computer vision**, achieving **90% accuracy**.
- Integrated the model into a **Django** backend server.
- Built a smart trash can prototype using an **ESP32-CAM** and **servomotors**.
- Integrated the prototype with the model via **REST API** and **Wi-Fi connection**, reaching a 3-second latency.
- Developed an interactive dashboard to visualize business metrics.

Technologies : ESP32, C/C++, Arduino, Tensorflow, Computer Vision, Django, Python

[PotAI](#)

IoT Smart Irrigation System with Machine Learning

github.com/adminLTR/PotAI

- Trained a **Random Forest regression model** to calculate the exact amount of water in milliliters that a plant requires.
- Achieved an **MAE of 0.02** and an **R2 coefficient of 0.98** during model training.
- Integrated the model into a **Flask** server.
- Built a smart planter prototype using an **ESP32**, soil and ambient humidity sensors, temperature sensors, and a water pump actuator.
- Integrated the prototype with the model via **REST API** and **Wi-Fi connection**.
- Developed a mobile application to control and monitor the prototype.

Technologies: ESP32, C/C++, Arduino, Scikit-learn, ML, Python, Flask

[CacaoLens](#)

Mobile application for automatic classification of cacao diseases using Deep Learning.

github.com/adminLTR/CacaoLens/releases/tag/v1.0.0

- Developed a mobile application in **Flutter** for the **visual classification** of cacao fruits into 3 categories.
- Built and integrated a **Deep Learning** model using **TensorFlow Lite**, running entirely on-device to execute offline inferences and reduce network dependency.
- Developed a **REST API** with **Node.js** and **Express.js** for user management, authentication, and predictions.
- Implemented data persistence using **PostgreSQL** and **Prisma ORM**.
- Achieved **93% accuracy** in classification, with an on-device inference time of approximately 300 ms.

Technologies: Flutter, Express.js, PostgreSQL, TensorFlow, Python, Flask, Docker, Git, Edge AI

WORK EXPERIENCE

[GM Group](#)

Fullstack developer

solucionesgm.com

Salto, Uruguay
Mar 2025 - Aug 2026

- Migrated from a monolithic architecture to **22 microservices**, enabling agile and modular scaling across 6 different systems.
- Automated business processes using **Artificial Intelligence** and **Web Scraping**, reducing manual workload by over 50%.
- Integrated external WhatsApp, SMS, and email services via **REST APIs** and **Webhooks**.
- Implemented modular and responsive web interfaces using **ReactJS** and **Tailwind CSS**.
- Restructured and optimized the **database**, eliminating duplicate records and redundant tables to improve data integrity.
- Implemented **Docker** to standardize and streamline the development environment setup.
- I worked with **Cursor** and **Copilot** with **Claude** models to make development easier with assisted AI

***Technologies:** React, Express.js, Node.js, Python, Flask, FastAPI, MySQL, API REST, TailwindCSS, Docker, Git, Selenium*

SKILL

- **IoT and Embedded Systems:** ESP32, ESP-IDF, Raspberry, Arduino, C/C++, sensors, actuators
- **Artificial Intelligence:** TensorFlow/Keras, LLM's, Scikit-learn, ML, DL, Edge AI
- **Backend:** Node.js, Express.js, Python, Flask, FastAPI, Django, PHP, Laravel
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB
- **Architecture and APIs:** Microservicios, MVC, API REST, Webhooks
- **Tools:** Jira, Git, Docker, Selenium, Web Scraping, Postman, Cursor, Copilot
- **Frontend:** ReactJS, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, SASS, jQuery, Flutter

EDUCATION

Universidad Nacional Mayor de San Marcos (UNMSM)

Software Engineering - Final year - *Top third*

Lima, Peru
Feb 2022 - Dec 2026

Universidade de São Paulo (USP)

Computer Science - Exchange student

São Paulo, Brazil
Jul 2026- Dec 2026

Alma Mater Studiorum - Università di Bologna (UNIBO)

Computer Engineering - Exchange student

Bologna, Italy
Feb 2026 - Jul 2026

LANGUAGES

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|--------------------|-------------------|---------------|
| • Spanish (Native) | • Italian(B2) | • French(A2) |
| • English (B2) | • Portuguese (B2) | • German (A2) |